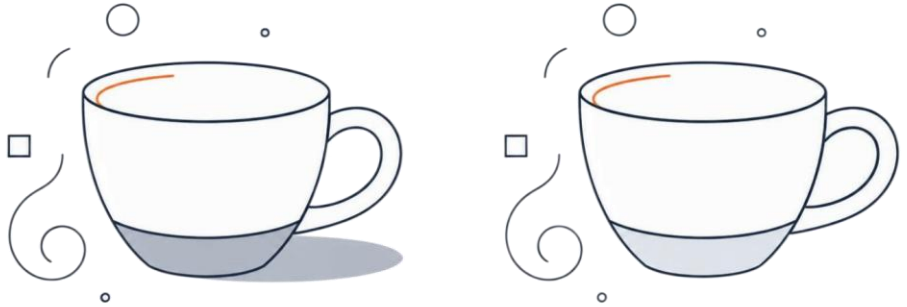
Point pixel processing

Changing each pixel using only its own value



Keep the picture’s geometry. Change how light or dark each pixel appears.

Start with grayscale: 0 is black, 255 is white.

The basic rule

One pixel in, one pixel out

A point operation reads a pixel's value and writes a new value at the same spot.

01

Read

Take the input intensity, r.

02

Transform

Apply the same rule to every pixel.

$$s = T(r)$$

03

Write

Store the output intensity, s, at that location.

Neighbors never matter, so the order of pixels is irrelevant.

Some Basic Gray Level Transformations

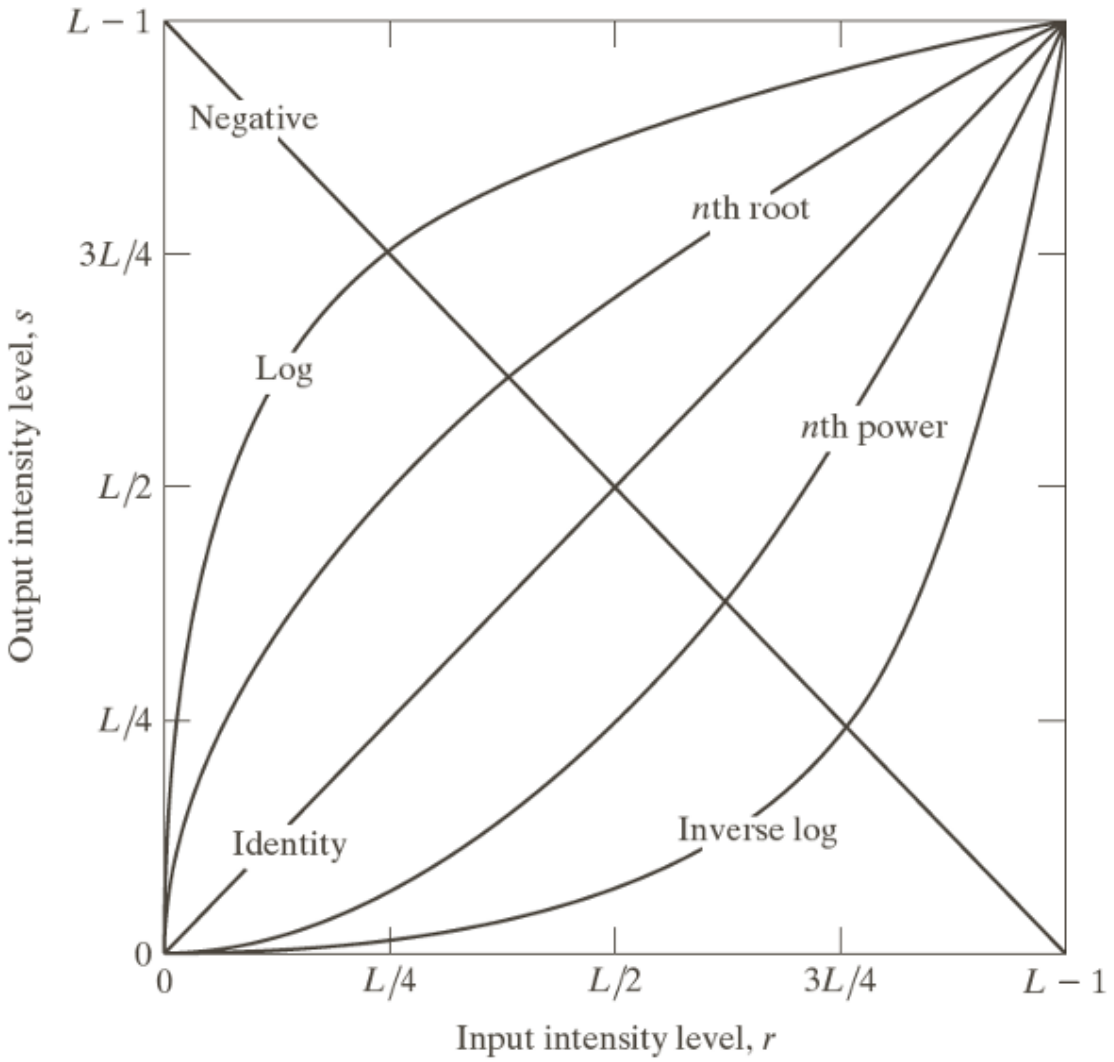
01 Identity

02 Inverse
S = 255-r

03 Brightness / Darkness
S = r ± c

04 Logarithmic
S = c * log(1+r)

05 Power-Law(Gamma)
S = c * (r^y)



Example 1

Q)A 4x4 , 4bits/pixel image passes through appoint-wise intensity transformation given by

$$S=T(r) = \alpha \log_2(1+r) + \beta$$

Where α and β are unknown parameters. Only a few pixels are available in the input and output images , as shown below

(a) Find α and β

(b)what is the value of the pixel with the "?" mark in the output image.

(c)what is the value of the pixel with the "?" mark in the input image.

	3		
		15	
	?		
1			3

T(r)

	5		
		11	
	8		
?			5

Example 1 sol

S=T(r) = $\alpha\log_2(1+r)+\beta$

a)

5 = $\alpha\log_2(1+3)+\beta$

5 = $2\alpha+\beta$

$\beta = 5 - 2\alpha$ ⟶ 1

11 = $\alpha\log_2(1+15)+\beta$

11 = $4\alpha+\beta$

Substitution from equation 1

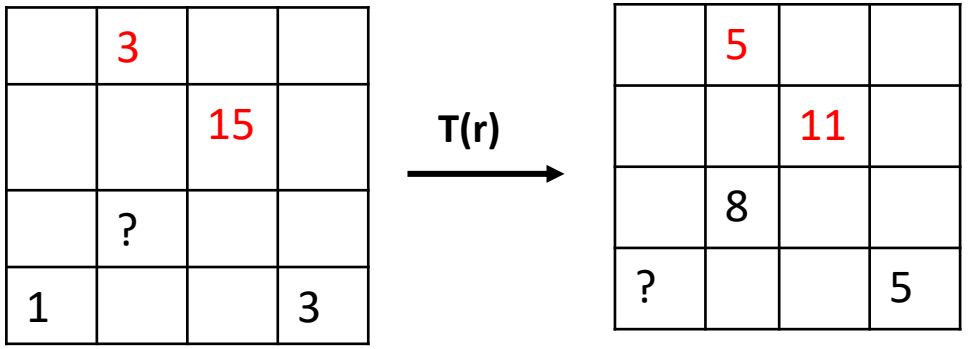
11 = $4\alpha + 5 - 2\alpha$

$6\alpha = 6$

$\alpha = 1$

Substitution in equation 1

$\beta = 5 - 2 = 3$



b)

Output = $\log_2(1+1)+3 = 4$

c)

8 = $\log_2(1+input)+3$

$\log_2(1+input) = 5$

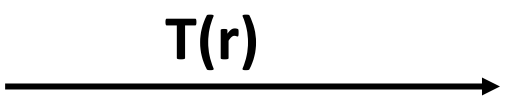
$1+input = 32$

$input = 31$

Example 2

Q) A4x4,4bits/pixel image (shown below) passes through one point-wise intensity transformations given by
S=T(r)= round($5\sqrt{r}$)

7	3	4	1
1	2	0	3
4	2	2	1
0	3	5	1



Example 2 sol

Q) A4x4,4bits/pixel image (shown below) passes through one point-wise intensity transformations given by
S=T(r)= round($5\sqrt{r}$)

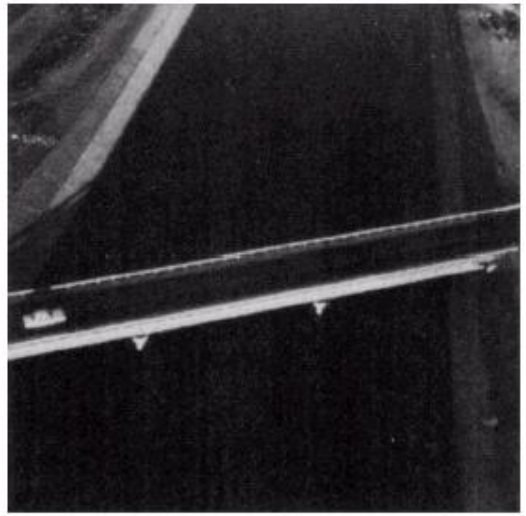
7	3	4	1
1	2	0	3
4	2	2	1
0	3	5	1

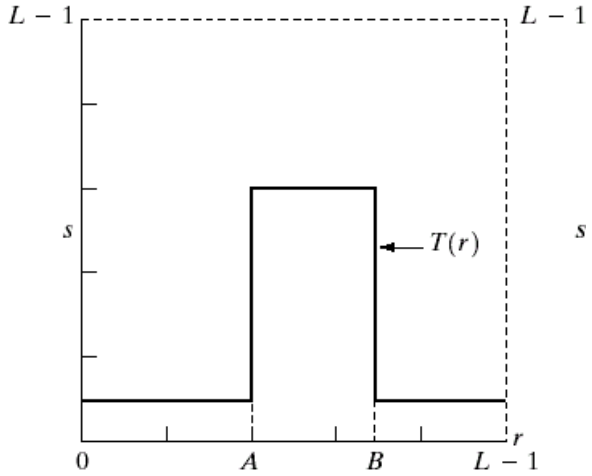
T(r) →

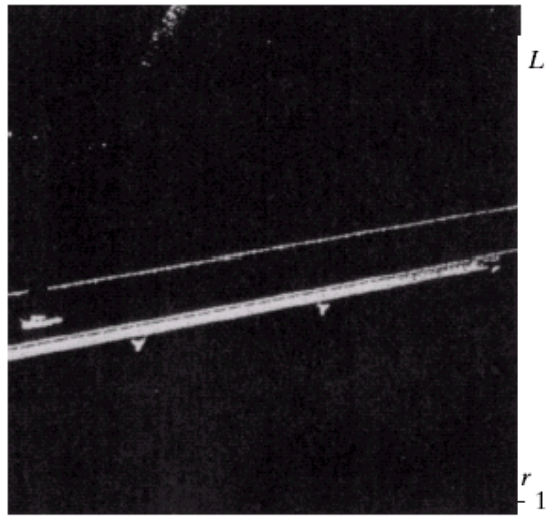
13	9	10	5
5	7	0	9
10	7	7	5
0	9	11	5

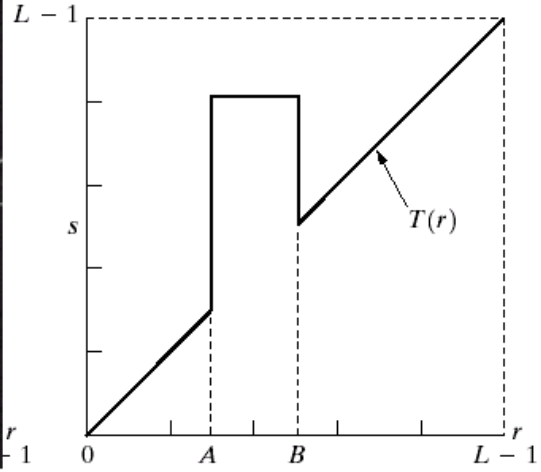
Intensity level slicing

- Highlights a specific range of grey levels.

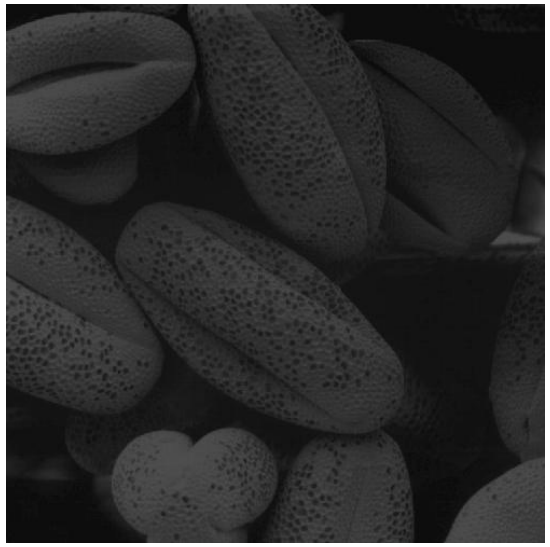
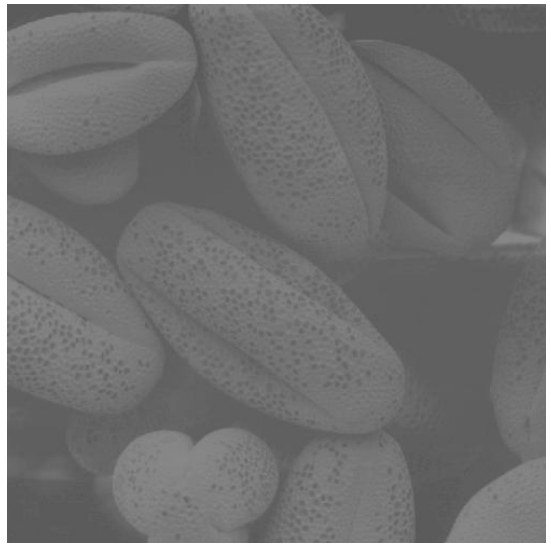








Contrast Stretching


Low Contrast

High Contrast

Contrast stretching

$$\text{New value} = \frac{value - oldmin}{oldmax - oldmin} * (new\ max - new\ min) + new\ min$$

Contrast Stretching Example

- Q) A4*4, 4bits/pixel original image is given by

6	13	12	13
12	6	7	12
13	7	7	12
14	11	11	14

Apply full-scale contrast stretch to the image . show your work and sketch the resulting image

- old min=6 old max=14
- new min=0 new max=15

- New value= $\frac{6-6}{14-6} * (15-0) + 0 = 0$
- New value= $\frac{13-6}{14-6} * (15-0) + 0 = 13$

0	13	11	13
11	0	2	11
13	2	2	11
15	9	9	15